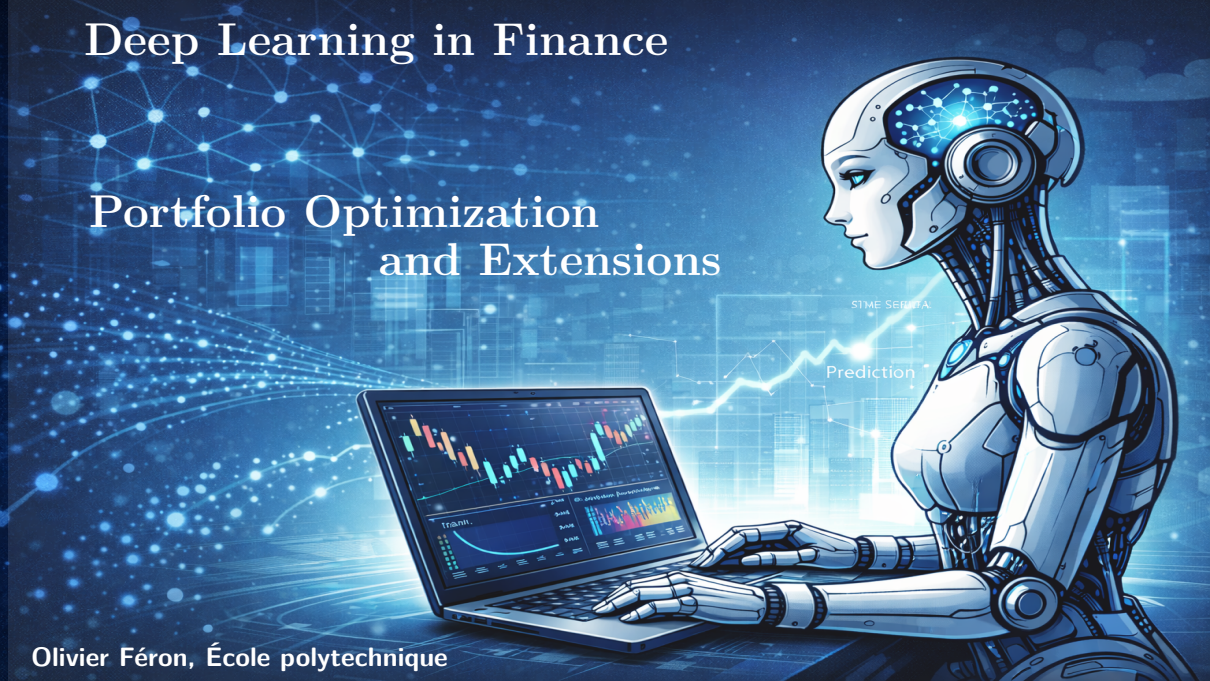
Deep Learning in Finance

Portfolio Optimization and Extensions



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2. Markowitz mean–variance
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5. Reinforcement Learning (conceptual overview)

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Objectives

- **Markowitz** mean–variance portfolio optimization, efficient frontier
- Extensions to other risk indicators: **VaR** and **CVaR / Expected Shortfall**
- **Neural networks** for fast approximation/parametrization of the efficient frontier
- **Reinforcement Learning** (conceptual): portfolio management as sequential decision making

From pricing to decision making

- Day 2–3: *Pricing* as function evaluation

$$V = F(x), \qquad x = (S_t, K, T, \xi)$$

- Day 4: *Portfolio choice* as optimization / control

$$\text{choose } \alpha \in \mathbb{R}^d \quad \Rightarrow \quad \text{trade-off return vs risk}$$

- Key question: **how to compute (and learn) optimal allocation policies efficiently?**

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Portfolio allocation

- We consider d risky assets first characterized by their price $S^i \in \mathbb{R}^+$, $i = 1, \ldots, d$.

Portfolio Allocation

A Portfolio allocation is a vector weight

$$\alpha \in \mathbb{R}^d$$

with the budget constraint:

$$\sum_{i=1}^{d} \alpha^i = \mathbf{1}^\top \alpha = 1$$

where each component α^i represents the fraction of total wealth invested in asset i.
When short selling is prohibited, the allocation further satisfies

$$\alpha^i \geq 0, \quad i = 1, \ldots, d,$$

in which case the investor fully allocates his wealth across the available assets

Portfolio Allocation, asset quantities and amount of money

- Given the total wealth W_t (in money unit !) at time t, the amount of money invested in asset i is then $\alpha_t^i W_t$, and the number of units held in asset i is

$$\varphi_t^i = \frac{\alpha^i W_t}{S_t^i}.$$

- The portfolio value at time t satisfies

$$W_t = \sum_{i=1}^{d} \varphi_t^i S_t^i,$$

  which ensures that the portfolio is self-financing.
- Unities:
  - W_t: total wealth in money unit (\$, €, ...)
  - S_t^i: price of asset i in money unit per unit of asset
  - φ_t^i: units of asset i held at time t
  - α^i: fraction of total wealth allocated to asset i, in percentage of wealth

Portfolio Return and Wealth Dynamics

- Consider the (arithmetic) returns of individual assets:

$$R_{t+1}^i = \frac{S_{t+1}^i - S_t^i}{S_t^i}.$$

- Then the portfolio return, with portfolio allocation α, is the weighted sum

$$R_{t+1}^\alpha = \alpha^\top R_{t+1} = \sum_{i=1}^d \alpha^i R_{t+1}^i.$$

- The associated wealth process follows

$$W_{t+1} = W_t(1 + R_{t+1}^\alpha) = W_t \left(1 + \sum_{i=1}^d \alpha^i R_{t+1}^i\right) = W_t(1 + \alpha^\top R_{t+1}).$$

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! The allocation α completely determines the portfolio dynamics This linear relation between asset returns and portfolio return is the foundation of mean–variance portfolio theory.

Portfolio optimization

- **Admissible set of allocations:**

$$\mathcal{A} = \left\{ \alpha \in \mathbb{R}^d : \mathbf{1}^\top \alpha = 1 \right\}.$$

- **The case of long only position:**

$$\mathcal{A}_+ = \left\{ \alpha \in \mathbb{R}^d : \mathbf{1}^\top \alpha = 1,\ \alpha^i \geq 0 \right\}.$$

- **Goal:** choose the "best" $\alpha \in \mathcal{A}$ or $\alpha \in \mathcal{A}_+$

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? What does it mean **?**

Setup and notation

- We define the mean and covariance of R:

$$\mu := \mathbb{E}[R] \in \mathbb{R}^d, \qquad \Sigma := \mathrm{Cov}(R) \in \mathcal{S}_+^d \tag{1}$$

  with $\mathcal{S}_+^d$ the set of symmetric definite positive matrices in $\mathbb{R}^{d \times d}$

- **Objective:=** maximize the expected portfolio return (performance) $\longrightarrow$ **solution:**
- **Objective:=** minimize the variance of the portfolio return (risk) $\longrightarrow$ **solution:**

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On Blackboard

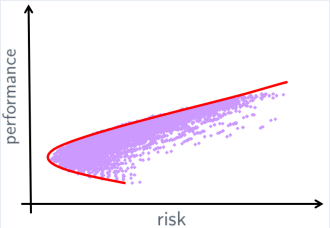
Mean–variance optimization, Efficient Frontier

! no portfolio has higher $\mathbb{E}[R^\alpha]$ with lower $\mathrm{Var}(R^\alpha)$

- We seek a tradeoff between performance (ex: expected return) and risk (ex: variance)

Efficient Frontier

The efficient frontier is the set of portfolios that are **Pareto-optimal**:



- If returns are Gaussian (or elliptical): mean–variance fully characterizes risk
- In practice: constraints, transaction costs, non-Gaussian tails $\Rightarrow$ go beyond variance

Mean–variance optimization, Efficient Frontier

Primal form

- **Objective:** Variance minimization at target expected return $\mu^\star$:

$$\min_{\alpha \in \mathbb{R}^d} \frac{1}{2}\alpha^\top \Sigma \alpha \quad \text{s.t.} \quad \begin{cases} \alpha^\top \mu = \mu^\star, \\ \mathbf{1}^\top \alpha = 1 \end{cases}$$

- Solution $\hat{\alpha}(\mu^*)$ and Efficient frontier obtained by varying $\mu^\star$

On Blackboard

Mean–variance optimization, Efficient Frontier

Regularized form

- **Risk-aversion (or regularized) form:**

$$\max_{\alpha} \; \alpha^\top \mu - \frac{\lambda}{2} \alpha^\top \Sigma \alpha \quad \text{s.t.} \quad \mathbf{1}^\top \alpha = 1$$

- $\lambda > 0$ controls the risk–return trade-off
- Convenient for computation: **a single objective, vary λ to trace the frontier**

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Quantile risk: Value-at-Risk (VaR)

Consider losses $L = -R^{\alpha}$ (positive values = losses)

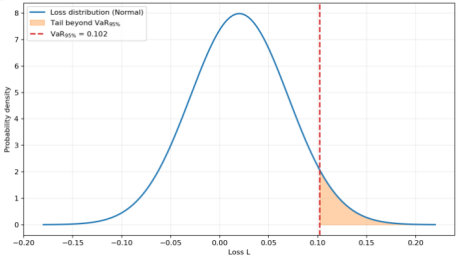
Value-at-Risk

For a confidence level $\xi \in (0,1)$, the Value-at-Risk (VaR) at level ξ is defined as

$$\mathrm{VaR}_{\xi}(L) := \inf\{\ell \in \mathbb{R} \ : \ \mathbb{P}(L \leq \ell) \geq \xi\}.$$

It is the smallest threshold ℓ such that the probability that the loss does not exceed ℓ is at least ξ. Equivalently, with probability $1-\xi$, the loss L exceeds $\mathrm{VaR}_{\xi}(L)$.

Although the Value-at-Risk is conceptually a monetary value of potential loss, it is mathematically more convenient to define it on returns (or losses derived from returns), and the two formulations are fully equivalent once multiplied by the portfolio value.



Tail risk: CVaR / Expected Shortfall

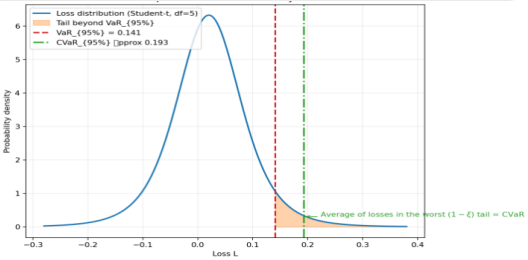
Conditional Value-at-Risk / Expected Shortfall

For a confidence level $\xi \in (0,1)$ and a loss variable L, the Conditional Value-at-Risk (CVaR), also called the Expected Shortfall, is defined as the expected loss conditional on exceeding the Value-at-Risk:

$$\mathrm{CVaR}_\xi(L) := \mathbb{E}[\, L \,|\, L \geq \mathrm{VaR}_\xi(L)] \,.$$

It represents the average loss in the worst $(1-\xi)$ fraction of cases, thus capturing the severity of tail events beyond the VaR threshold.

- Better tail sensitivity; convex optimization formulations (under scenarios)
- In practice: scenario-based optimization $\Rightarrow$ computational bottleneck at scale



From Risk Measures to Machine Learning: The Estimation Problem

Pdf estimation

All portfolio optimization methods — whether based on mean–variance, VaR, or CVaR — rely on an underlying distribution of asset returns. In practice, this means:

- estimating at least the first two moments $(\hat{\mu}, \hat{\Sigma})$ in the Gaussian setting;
- or generating scenarios that approximate the joint probability distribution of returns.

Practical limitations

Financial data exhibit:

- very low signal-to-noise ratio $\rightarrow$ unstable estimates of $\hat{\mu}$;
- non-stationarity and regime shifts $\rightarrow$ parameters vary over time;
- fat tails and dependence structures poorly captured by simple (Gaussian) models.

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Two ML/NN viewpoints

1. **Approximate the efficient frontier with a Surrogate**

$$(\mu, \Sigma, \lambda) \mapsto \alpha_\theta(\lambda)$$

2. **Direct learning by optimization** (differentiable objective):

$$\min_\theta \ \mathbb{E}\Big[\underbrace{\alpha_\theta^\top \Sigma \alpha_\theta}_{\text{risk}} - \lambda \underbrace{\alpha_\theta^\top \mu}_{\text{return}}\Big]$$

Approach 1: Learning the efficient frontier

- Instead of solving N_λ optimizations, learn the efficient frontier $\alpha_\theta(\lambda)$ once
- Sweep λ on a grid and generate:

$$(\sigma_p(\lambda),\ \mu_p(\lambda)) = \left(\sqrt{\alpha_\theta(\lambda)^\top \Sigma \alpha_\theta(\lambda)},\ \alpha_\theta(\lambda)^\top \mu\right)$$

- Practical advantage: **fast re-computation** for stress tests / rolling windows

Constraint handling: simplex parametrization

- Long-only + budget: $\alpha^i \geq 0$, $\sum_i \alpha^i = 1$
- Parametrize with softmax:

$$\alpha_\theta = \mathrm{softmax}(z_\theta) \qquad (z_\theta \in \mathbb{R}^d)$$

- Extensions:
  - allow shorting: $\alpha = \tanh(z)/\mathbf{1}^\top |\tanh(z)|$
  - leverage cap: projection onto $\{\alpha : \|\alpha\|_1 \leq L\}$

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From static optimization to sequential decisions

- Markowitz: one-shot allocation $\Rightarrow$ static
- Real trading: rebalancing, costs, risk constraints, evolving signals
- Portfolio management as **control problem**:

$$\text{state } s_t \;\rightarrow\; \text{action } a_t \;\rightarrow\; \text{reward } r_t$$

RL ingredients for portfolio management

- State s_t: prices, returns, vol, technical indicators, macro features
- Action a_t: target weights w_t or trades Δw_t
- Reward r_t (examples):

$$r_t = \underbrace{R_{p,t}}_{\text{PnL}} - \eta \underbrace{\mathrm{Risk}_t}_{\text{penalty}}; \qquad r_t = \log(1 + R_{p,t}) - c\|\Delta w_t\|_1$$

- Policy $\pi_\theta(a|s)$ learned to maximize expected cumulative reward

Wrap-up / takeaways

- Markowitz provides a clean mathematical baseline (risk–return trade-off)
- VaR/CVaR: capture tail risk but increase numerical complexity
- DL can:
  - approximate the efficient frontier / optimizer map
  - handle constraints via parametrization
  - improve speed in repeated computations (rolling windows, stress tests)
- RL: natural next step for dynamic portfolio decisions (costs, constraints, multi-period)